Robust Data-Driven Output Feedback Control via Bootstrapped Multiplicative Noise

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Problem Setting

Linear Time-Invariant System $\quad x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{D}_w(0, W)$
$\quad y_t = Cx_t + v_t, \quad v_t \sim \mathcal{D}_v(0, V)$

Input-Output Dataset $\quad \mathcal{D}_{\mathcal{T}} = \left(y_{0:\mathcal{T}}^{\text{train}}, u_{0:\mathcal{T}}^{\text{train}}\right)$

History-dependent Feedback Policy $\quad u_t = \pi(y_{0:t}, u_{0:t-1})$

Linear Dynamic Feedback Policy $\quad \hat{x}_{t+1} = F\hat{x}_t + Ly_t$, $\quad u_t = K\hat{x}_t$

Infinite-horizon Quadratic Performance Criterion $\quad J(\pi) = \lim_{\mathcal{T}\to\infty} \frac{1}{\mathcal{T}} \mathbb{E} \sum_{t=0}^{\mathcal{T}-1} (x_t^\intercal Q x_t + u_t^\intercal R u_t)$

Time-$\mathcal{T}$ and Optimal Performance Criterion $\quad J_{\mathcal{T}} = J(\pi_{\mathcal{T}})$, $\quad J^* = \min_{\pi} J(\pi)$

Goal

A control architecture that achieves a distribution of $J_{\mathcal{T}}/J^*$ with *small mean and small tails*, especially in the small-data regime

Obstruction

True system (A, B, C, W, V) is not known beforehand…

…only a *finite, noisy dataset $\mathcal{D}_{\mathcal{T}}$* is available

Sequential Model-based Design Paradigm

Estimate a model from data → Design a control policy based on model

Existing Approaches

Certainty Equivalence
- Ignores model uncertainty altogether
- Can lead to very poor tail performance – *fragile*

Isotropic Robustness
- Fails to size and shape uncertainty accurately
- Can lead to very poor average performance – *inept*

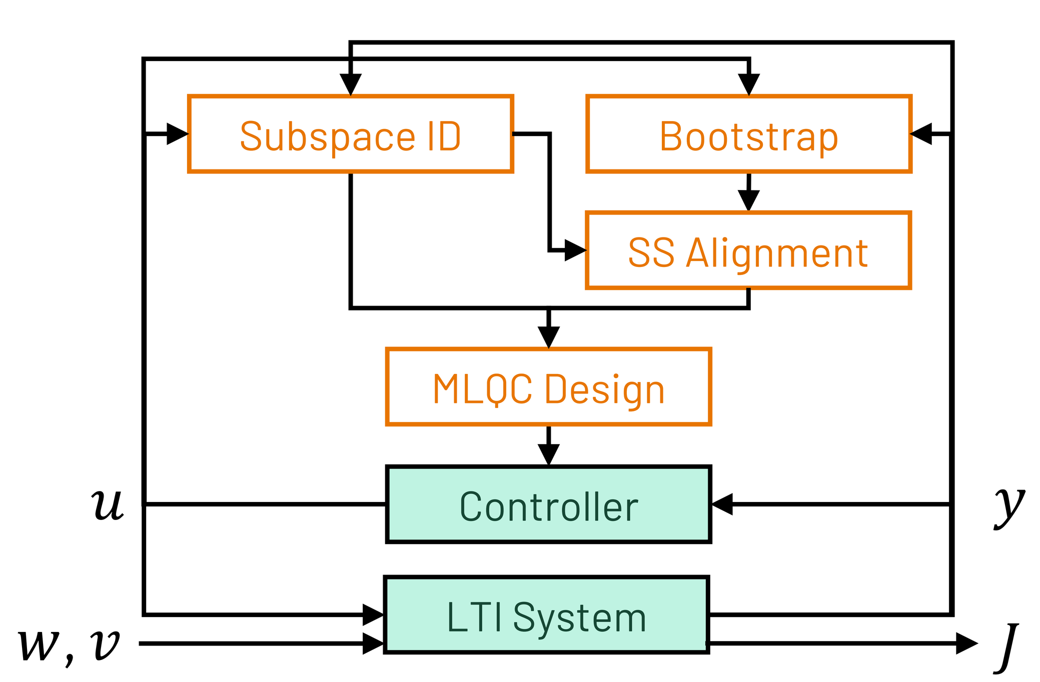
Control Architecture

Main Idea

Share a stochastic uncertainty representation between the model uncertainty description and the robust control design method

Uncertainty size and shape are aligned between components

Extend [1] from the state-feedback to the output-feedback setting

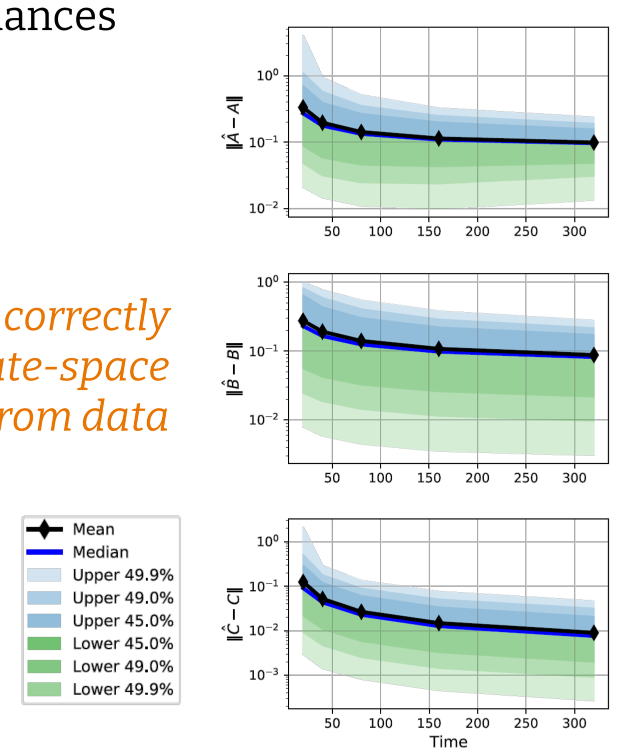


Component Details

Subspace Identification

Use *N4SID subspace identification* algorithm to estimate a state-space model
1. Construct block Hankel matrices from observed input-output data
2. Estimate subspaces via singular value decomposition
3. Retrieve system matrices and additive noise covariances

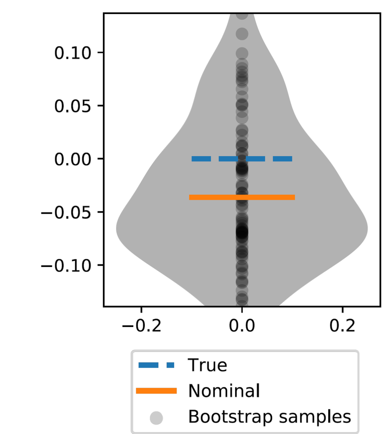
Subspace ID correctly identifies state-space models from data



Bootstrap Uncertainty Estimation

Use *semiparametric bootstrap* to preserve temporal dependence in time-series data:
1. Draw iid random samples with replacement from a finite dataset of empirical process & measurement noise from residuals calculated with the nominal system
2. Construct lots of fake datasets of same size as original
3. Use subspace ID to compute system matrices for each fake dataset
4. Use bootstrap distribution to estimate covariance of system matrices

Bootstrap distribution captures variation between true system and estimated system



State-space Alignment

Due to *non-uniqueness* of state space representations, the uncertainty representation should not be obtained directly from a sample covariance of the bootstrap resamples.

For each resample we find a *similarity transformation* that minimizes squared error to the nominal state space model, then compute a sample covariance in the transformed coordinates.

Linear matrix equation with analytic solution $\quad \min_{T \in \mathbb{R}^{n \times n}} \psi_A \|T\bar{A} - \hat{A}T\|_F^2 + \psi_B \|T\bar{B} - \hat{B}\|_F^2 + \psi_C \|\bar{C} - \hat{C}T\|_F^2$

Multiplicative Noise Linear Quadratic Control

MLQC Optimal Control Problem

$$\begin{aligned} \underset{\pi \in \Pi}{\text{minimize}} \quad & \lim_{\mathcal{T}\to\infty} \frac{1}{\mathcal{T}} \mathbb{E} \sum_{t=0}^{\mathcal{T}-1} (x_t^\intercal Q x_t + u_t^\intercal R u_t) \\ \text{subject to} \quad & x_{t+1} = (A + \bar{A}_t)x_t + (B + \bar{B}_t)u_t + w_t \\ & y_t = (C + \bar{C}_t)x_t + v_t \end{aligned}$$

$\bar{A}_t, \bar{B}_t, \bar{C}_t$ are iid zero-mean random matrices whose vectorizations have covariances $\Sigma_A, \Sigma_B, \Sigma_C$, which are taken from the bootstrap uncertainty

A, B, C, W, V, U estimated with subspace ID

Q, R penalties chosen & known by designer

Policy class $\Pi \leftarrow$ linear dynamic output feedback

No separation between estimation and control, optimal gains (F, K, L) must be *jointly computed*

MLQC Solution

$$(\underbrace{X}_{4\times\ \text{psd}}, \underbrace{F, K, L}_{\text{Gains}}) = \text{GDARE}(\underbrace{A, B, C, W, V, U, Q, R, \Sigma_A, \Sigma_B, \Sigma_C}_{\text{Problem data}})$$

Four coupled generalized Riccati equations

Solved via a value-iteration algorithm [2]

Solution depends on noise covariances

When covariances are too large, bisection used to scale noise to the "uncertainty threshold"

MLQC policy *induces robustness* towards the multiplicative noise directions (this effect is well-known in the literature)

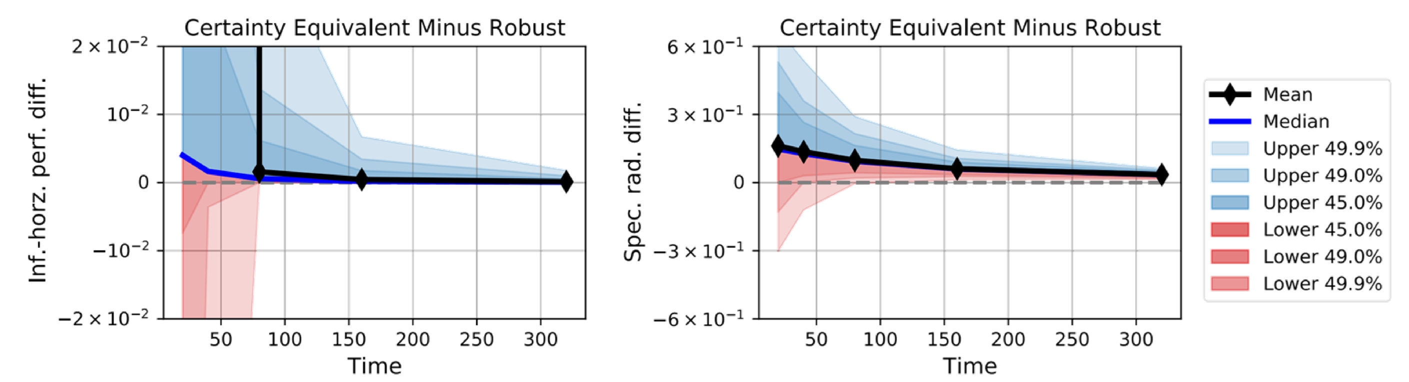
Numerical Experiments

2-state shift-register system, adapted from [3]

$$\begin{bmatrix} A & B \\ C & \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & \end{bmatrix}, \quad \begin{bmatrix} Q & \\ & R \end{bmatrix} = \begin{bmatrix} 1 & -1 & \\ -1 & 1 & \\ & & 0.01 \end{bmatrix}, \quad \begin{bmatrix} W & \\ & V \end{bmatrix} = \begin{bmatrix} 0.1 & 0 & \\ 0 & 0.1 & \\ & & 0.1 \end{bmatrix}$$

Extremely sensitive to model identification errors

Any small error in the estimated system matrices produce an unstable closed-loop system when using LQG control – *demands robustness*



Difference between certainty-equivalent (CE) (naïve) and robust multiplicative noise-based (RMN) (proposed) control on infinite-horizon performance $J_{\mathcal{T}}/J^*$ and closed loop spectral radius $\rho(\Phi_{\mathcal{T}})$ vs time.

RMN control tends to produce highly stable controllers, limiting state growth and *mitigating the risk of poor performance*

Conclusions

We proposed a data-driven control architecture that uses unconventional multiplicative noise-based policy synthesis to explicitly account for bootstrap-estimated model uncertainty

Numerical experiments verify enhanced risk sensitivity

Future Work

- Provide finite-time theoretical performance guarantees
- Explore alternative uncertainty quantification schemes
- Explore alternative robust control synthesis frameworks

Acknowledgements

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Paper	https://arxiv.org/abs/2205.05119
Code	https://github.com/TSummersLab/robust-adaptive-control-multinoise-output
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[2] W. L. De Koning, *"Compensatability and optimal compensation of systems with white parameters,"* IEEE Transactions on Automatic Control, 37(5):579–588, 1992.
[3] Benjamin Recht, *"Digital Witnesses,"* 2020. https://www.argmin.net/2020/07/27/discrete-fragility/